

How to Communicate

Tao LIN

August 31, 2026



- 1 A General Guide
 - Why Communication Matters?
 - The 7 **C**'s of Communication
- 2 How to Communicate With Your Collaborator?
 - How to Work With Your Advisor Effectively
 - How to Share Progress With Your Mentors/Collaborators?
 - How to Work With a Busy Advisor?
 - How to Work With Your Senior Advisor(s)?
- 3 How to Ask Questions The Smart Way (From CS Perspective)?
 - Before You Ask
 - When You Ask
- 4 How to Do Presentation

Course Schedule

Week	Date	Topics
1	2026. Sep. 01	How to communicate
2	2026. Sep. 08	How to do presentation
3	2026. Sep. 15	How to be a good AI researcher (I): doing research I
4	2026. Sep. 22	How to be a good AI researcher (II): productivity and career
5	2026. Sep. 29	How to be a good AI researcher (III): academic paper writing and peer reviews
6	2026. Oct. 13	Sharing the experience of writing excellent academic papers and rebuttal
7	2026. Oct. 20	Practice course I
8	2026. Oct. 27	Practice course II

Acknowledgement

- [The 7 Cs of Communication](#), World of Work Project
- [How To Ask Questions The Smart Way](#), Eric Steven Raymond
- [Awesome Tips](#), JiaBin Huang

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Why Good Research Gets Stuck

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Why is this hard to answer?

- The goal and expected behavior are unclear.
- The setup, evidence, and attempted diagnosis are missing.
- The reader cannot tell what decision or help is needed.

Why Good Research Gets Stuck

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Communication creates shared understanding for a decision.

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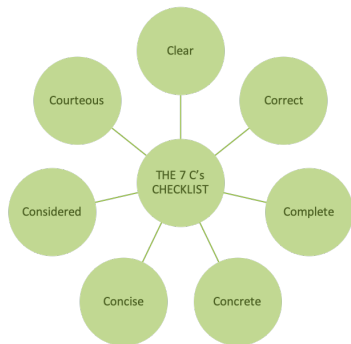
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The 7 Cs Turn a Message into Useful Action



The 7 Cs are not seven isolated writing tips. They test whether a message is

- understandable,
- trustworthy,
- efficient, and
- actionable for this audience.

Clear: State the Purpose and the Ask

- The reader should immediately know **why** they received the message.
- Put the conclusion or decision point before supporting detail.
- Separate multiple goals instead of hiding them in one paragraph.

Research rule: Start with “I am sharing X because we need to decide Y.”

Case: Can the Reader Find the Point?

Dear Area Chairs and Senior Area Chairs,

We extend our sincere gratitude for your valuable efforts in the review process.

In our rebuttal, we have: (i) Clarified our contributions; (ii) Thoroughly reviewed and rephrased the manuscript for improved readability; (iii) Added additional experiments to verify effectiveness, including comparisons with more baselines across various datasets and tasks in attached PDF; (iv) Expanded discussions on related works.

We believe we have adequately addressed all the reviewers' concerns:

- Common concerns regarding logical expression and notation have been addressed through manuscript revisions (ii).
- Reviewer fjn's request for more extensive verifications through larger scale experiments has been met (iii).
- Reviewer K3hg had significant misunderstandings regarding some techniques in related works and misinterpreted some of our technical solutions and theoretical results. For instance, they incorrectly claimed that RDED [a] selects hard samples with a larger classification loss, arguing that this contradicts our theoretical results. In fact, RDED selects easy samples with a smaller loss, which is consistent with our analysis. We have clarified these misunderstandings and explained our theoretical results and technical solutions in detail (i and iv).
- Reviewer zbwg's requests for experiments on high-resolution datasets, more downstream tasks, and additional SOTA comparisons have been addressed (iii). We have also clarified the generalization of our theoretical results (i).
- Reviewer 7u7z's request for more detailed discussions about the relationship between our contributions and related work has been thoroughly addressed (i and iv).

We appreciate your time and effort. We believe our paper is now significantly strengthened after the rebuttal and hope this message assists in your decision-making process.

Best regards,

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Best regards,

Diagnosis: purpose first, response categories visible, evidence grouped under each claim.

Correct: Credibility Depends on Details



Before sending, verify

- numbers and units,
- experimental conditions,
- citations and links,
- labels, terminology, and grammar.

One visible error makes every other claim harder to trust.

Complete: Supply What the Decision Requires

Hard to answer

“Training is unstable. What should I do?”

Complete enough to diagnose

- goal and expected behavior,
- model, data, and environment,
- what changed,
- evidence and attempted fixes,
- the decision or help needed.

Complete does not mean exhaustive; it means **sufficient for this audience and decision.**

Concrete: Replace Impressions with Evidence

Vague

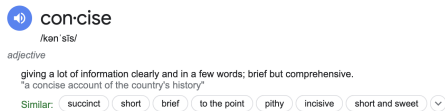
“The new method performs much better and is more stable.”

Concrete

“Accuracy increased from 81.4 to 84.2 across five seeds; standard deviation fell from 1.8 to 0.7.”

Use numbers, examples, logs, figures, and credible sources that the reader can inspect.

Concise: Preserve the Signal



A concise research update prioritizes

- 1 what changed,
- 2 what the evidence means,
- 3 what happens next.

Put supporting details behind links or in an appendix.

Concise means **high information value**, not merely fewer words.

Courteous: Protect the Working Relationship



Courtesy does not require agreement.

- Critique the claim, not the person.
- Acknowledge useful evidence before disagreeing.
- Make the next exchange easier, not harder.
- Stay professional when the other message is not.

Case: When Does Firm Become Escalatory?

Dear ACs and SACs:

We greatly appreciate the feedback from all reviewers and the AC, and we are committed to addressing concerns to improve our work. However, we must express our deep concern regarding Reviewer tGWA's evaluation, which appears **biased and ad-hoc, potentially hindering the development of community**. Our specific concerns are detailed below:

A. Fundamental Misunderstanding of Our Approach

Reviewer tGWA **repeatedly** mischaracterized our approach as text/LLM-based, despite our clear and consistent indication that our work focuses on a multimodal LMM (Large multimodal model).

B. Unreasonable Expectations Beyond Medical Theory

The reviewer criticized our method for not achieving tasks that are “not visible to the human eye.” **Such a request fundamentally contradicts the basis of medical theory and is both unreasonable and perplexing.** AI in pathology is designed to replicate the work of human experts to assist pathologists, not to surpass human diagnostic visibility.

C. Shifting Standards and Misaligned Comparisons

Initially, the reviewer demanded comparisons with models that are **four times larger and trained on 800 times more data**. After we conducted additional experiments demonstrating our method's effectiveness, the reviewer **shifted the focus to criticize our lack of genetic diagnostic capabilities** and minimized the significance of our performance improvements over much larger models.

D. Misrepresentation of Data Sharing and Usage

Despite our open release of data and models, the reviewer unfairly criticized us for not sharing data processing steps and falsely claimed that we only demonstrated how to use open_clip. In reality, the instructions provided clearly guide researchers on how to load our pre-trained model (see the historical version of our GitHub link from two weeks ago: [GitHub link](#)).

E. Ever-changing Concerns

Even after we addressed all the reviewer's concerns—including data, models, processing steps, comparisons, and additional experiments—the **reviewer did not adjust their evaluation score. This raises the question of whether the original concerns were genuinely the basis for the clear rejection decision.**

Given the severe bias evident in Reviewer tGWA's evaluation, we respectfully and sincerely request the AC to carefully reconsider the this reviewer's evaluation when making the final decision on our work.

Best regards,

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Best regards,

Diagnose it: Which sentences preserve respect, and which ones make collaboration harder?

Case Repair: Disagree Without Escalating

Thank you for your insights. To the best of our knowledge, models like [HIPT](#) do not perform mutation prediction, and the literature on these models does not claim that their performance on molecular mutation tasks surpasses that of human experts. If there are specific instances where this is indeed demonstrated, we would be eager to learn about them. Additionally, the paper [Campanella et al. 2024](#) you mentioned also noted that "it may be unknown whether a particular genomic alteration leads to a measurable morphological change visible in H&E stained slides."

Ultimately, it might not be necessary to focus too much on this specific point. Whether as reviewers or authors, our collective goal is to build a better AI community that serves society. No single work can solve all challenges at once; we all stand on the shoulders of giants, making small, incremental improvements that eventually lead to significant breakthroughs. This is our shared mission.

While we may have differing viewpoints today, we appreciate your feedback because we believe that discussions like these are an essential part of the journey towards AI eventually achieving this goal.

Separate **evidence**, **interpretation**, and **request**. Remove claims about motives or competence.

Considered & Coherent: Design for This Reader

- Restore the context that this audience is unlikely to remember.
- Match the level of abstraction to their role and expertise.
- Order claims so that each one prepares the reader for the next.
- Keep terminology, tone, and recommendations consistent across messages.

The same result needs a different explanation for a lab mate, an advisor, and a reviewer.

Use the 7 Cs to Build a Research Update

Clear

State the goal, conclusion, and decision needed.

Correct + Concrete

Report verifiable observations, conditions, and evidence.

Complete

Include enough context to reproduce the reasoning.

Concise

Prioritize what changed, what it means, and what happens next.

Courteous

Respect the reader while disagreeing or reporting failure.

Considered

Adapt the abstraction level and preserve coherent logical flow.

The rest of this lecture applies the same principles to collaborators, meetings, and technical questions.

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Collaboration Happens Under Limited Context and Attention

Your collaborator may

- have only 30–60 minutes for the meeting,
- follow several projects at different levels of detail,
- know the fundamentals but not every recent tool or experiment,
- provide better feedback when the reasoning and decision are visible.

The 7 Cs reduce the cost of restoring context and making a decision.

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Treat Advising as an Iterative Feedback Loop



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- ✗ Instruction only: the student waits instead of developing judgment.



Treat Advising as an Iterative Feedback Loop

- ✗ Final-result only: feedback arrives too late to change the process.
- ✗ Instruction only: the student waits instead of developing judgment.
- ✓ Iterative exchange: evidence and interpretation produce timely guidance.



How to work with your advisor effectively: show your work

How do you get the best guidance from your advisor?

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- the detailed process you went through,
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- the methodology you adopted,
- and the interpretations of the results you got.

How to work with your advisor effectively: Don't try figuring everything out yourself

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- BUT, If you spend 15 mins googling and still don't know where to start,
- please reach out to your peers/mentors.
- Asking for help is not a sign of weakness.

How to work with your advisor effectively: Frequent update

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- **After**: Send a summary and an actionable plan to keep everyone on the same page.

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- NO!
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- Discuss the problems with your mentors/collaborators.
- Help them help you get unstuck.

How to work with your advisor effectively: One single slide deck

- Put ALL the progress/results/figures/discussions in one single slide deck.

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- This saves **5 mins** in the meeting locating files and trying to retrieve results two weeks ago when someone asks for it.

How to work with your advisor: Communicate at the right level of abstraction

Facts:

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Suggestions:

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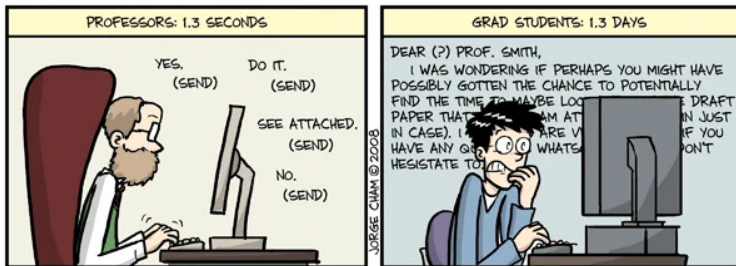
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AVERAGE TIME SPENT COMPOSING ONE E-MAIL



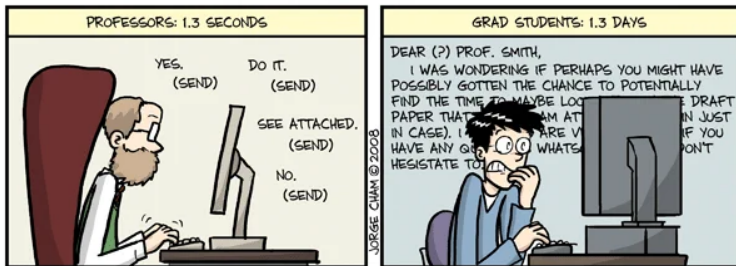
WWW.PHDCOMICS.COM

How to work with your advisor: leverage async discussions

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Keep your advisor engaged and excited about your research.

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- One topic at a time

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 - ✓ Embed the content inline.

Case: Structure Makes an Async Update Scannable

Thread

Siyuan Today at 10:28 AM
Hey, Prof. @Tao Lin, About our motivation figure, do you think we should highlight the novel setting or the mechanisms (such as metrics) ?

8 replies

Tao Lin Today at 10:34 AM
where is the motivation figure?

Jiaqi Shao Today at 10:45 AM
The motivation figure will be placed in the introduction section.
We are discussing about the "motivation":
1. [Novel setting for flexible MAS] highlight the flexibility of decentralized setting (general setting). e.g. given the evolving task, the MAS with predefined profiles/structures cannot effectively adapt to this task.
2. [Profile update for **efficient** decentralized MAS] highlight the rationale of designing "three" typical metrics for profile update, e.g. without using these metrics, the agents tend to pay much more time for profile/task allocation which is inefficient.

Tao Lin Today at 10:49 AM
ah, i thought we have already created a motivation figure

Jiaqi Shao Today at 10:51 AM
Not yet. We may consider both?
These two motivations also correspond to our two contribution. 1) novel decentralized setting. 2) profile update design.

Tao Lin Today at 10:58 AM
or try to highlight the limitation of current design, to motivate why do we need a novel setting for flexible MAS?
for 2), it can be moved to the methodology part

Siyuan Today at 11:00 AM
make sense

How to work with your advisor: Asynchronous communication II

- Async sharing before sync meeting

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- Agenda, agenda, agenda
 - ✗ Let's hop on a call and chat next week.
 - ✓ Send a clear meeting agenda with allocated time.

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A result that does not match the hypothesis can still produce useful evidence.

How could we share these “failed results”
and have productive conversations with your mentors/collaborators?

A Research Update Operationalizes the 7 Cs

- **Design:** Why this experiment? [Complete, Considered]
- **Hypothesis:** What should happen? [Clear]
- **Observation:** What happened? [Correct, Concrete]
- **Interpretation:** What does the evidence imply? [Coherent]
- **Visualization:** What makes the pattern visible? [Concise]
- **Next step / ask:** What should happen now? [Clear, Considered]
- **Alignment:** Follow the plan, or justify a revision. [Courteous, Coherent]

How to share progress (Design: Why do we want to do this experiment?)

- Do not assume that collaborators retain every detail between meetings.

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- Remind them WHY you ran a particular experiment or implemented a particular component.

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- Do not assume that collaborators retain every detail between meetings.
- Remind them WHY you ran a particular experiment or implemented a particular component.
- This will provide the context for them to help interpret the results and steer the direction of your research.

How to share progress (Hypothesis: What do we expect to see?)

- Before showing your results,

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- Before showing your results,
- comment on what should have happened (if everything is correct)?

How to share progress (Observation: What did we see?)

- Show the (failed) results.

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- Show the (failed) results.
- Don't just say "It doesn't work."

How to share progress (Observation: What did we see?)

- Show the (failed) results.
- Don't just say "It doesn't work."
- Describe HOW it fails (with details and ideally in a self-contained manner).

How to share progress (Interpretation: Is this expected/working? please no “It doesn’t work.”)

After showing your results, comment on how the results align with or deviate from your expectations.

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 - I’ve narrowed down the problem to step B .
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 - Here is HOW it fails.
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 - I believe the core issues lie in steps Z and W .
 - I have ruled out W as the cause.
 - Next, I will design experiments to isolate the step Z .

How to share progress (Visualization: Any better ways to see the results?)

Seeing the results with a good visualization helps

- deepen our understanding

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 - you get to understand why specific feedback was given.

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- Once everyone agrees on an actionable plan, use it as the default next step.

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- Once everyone agrees on an actionable plan, use it as the default next step.
- Quite often junior students may go ahead and work on some other tasks instead.
- If new evidence changes the plan, explain the evidence and agree on a revised next step.

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So what should we do?

- Collaborate with your peers or senior students
 - Feel free to reach out to other students in your lab (especially if you share similar interests).
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 - Ask your advisor if you can continue collaboration with them.
- Try ad hoc meetings
 - Try to find a few minutes to meet with your advisor after their class or during office hours
 - Be prepared, concise, and respectful of their time.

- Make them excited about your work
 - Share frequent updates on your progress or exciting findings.
 - Show enthusiasm and make them excited!

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- Explore different advisors or co-advisors?
 - If working with your current advisor is consistently challenging, consider exploring other advisors who align better with your research interests.

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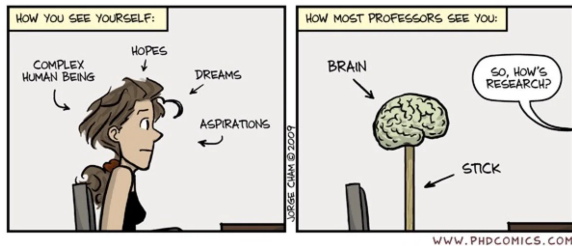
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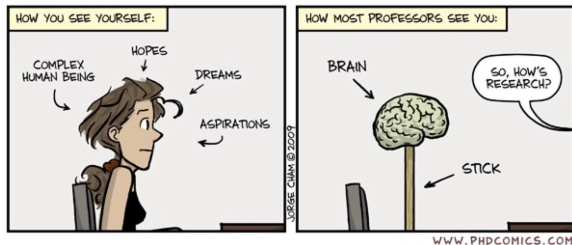
Check out below for some similar tips.

Pre-process your input



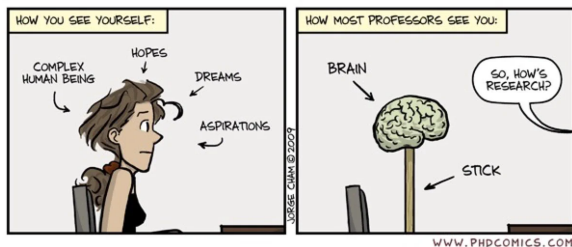
Pre-process your input

- Feedback quality depends on the context and evidence you provide.



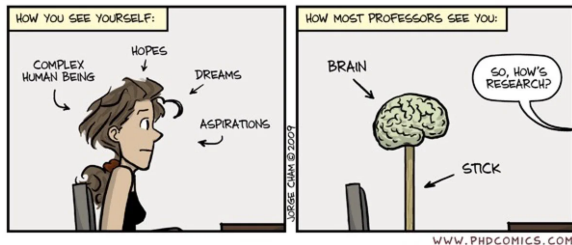
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Pre-process your input

- Feedback quality depends on the context and evidence you provide.
- Senior professors won't keep track of all the latest papers. But they sure know the fundamentals.
- Pre-process/abstract/simplify your work so that they can give you great feedback.



Post-process their output

- Senior professors may have deep insights to your research problem. But, they don't have the modern toolboxes you are familiar with.

Post-process their output

- Senior professors may have deep insights to your research problem. But, they don't have the modern toolboxes you are familiar with.
- Instead of taking their suggestions as is (e.g., implement some heuristics), map them into modern frameworks.

Find hands-on collaborators

When you are just getting started on your first project, make sure to find hands-on collaborators (other assistant prof, post-doc, or senior students in the lab).

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When you are just getting started on your first project, make sure to find hands-on collaborators (other assistant prof, post-doc, or senior students in the lab).

You will learn valuable skills from them!

Be transparent

- Senior professors are busy. Sometimes you may not get to interact with them for months!

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 - what have you done?

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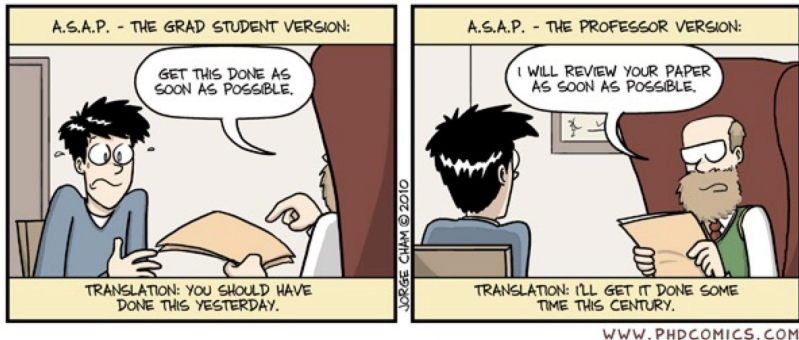
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 - what you will do?
 - when you will be on vacation?

Be specific

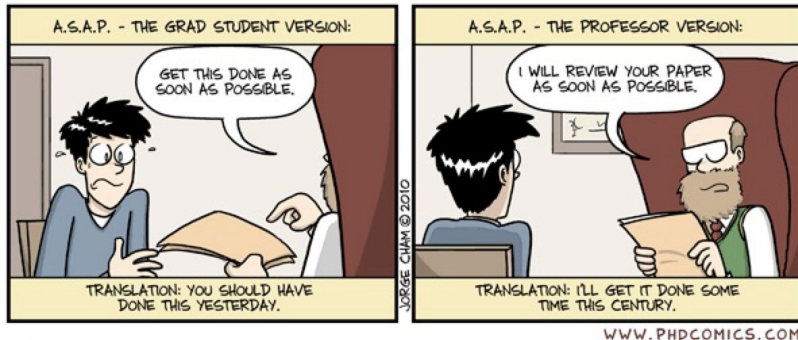
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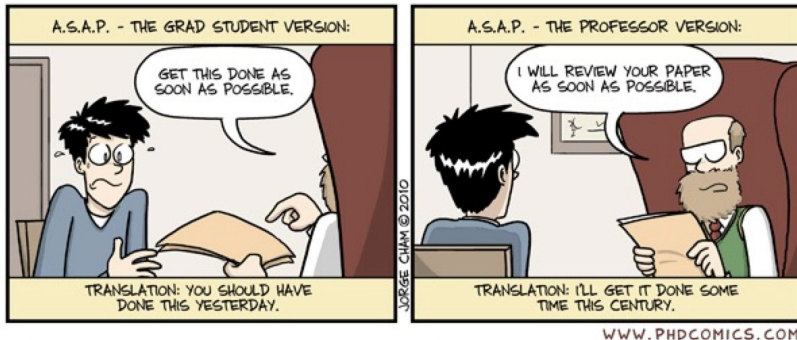
- ✓ Helps your advisor include this task in their to-do.



Be specific

Follow up with your professor's "I will review your paper soon." and ask for a specific date.

- ✓ Helps your advisor include this task in their to-do.
- ✓ You get to know when to follow up again.



Share progress early and frequently

Emphasize it again!

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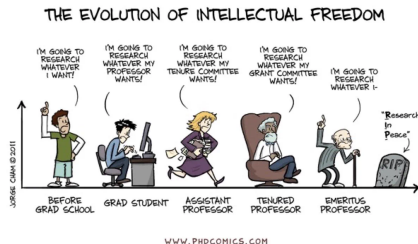
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- Many students feel intimidated about sharing results that are “not ready”.
- It often leads to a vicious cycle of “not ready” -> “no feedback” -> “build up more stress”.
- Break that cycle and keep engaging with your advisor.

Emphasize it again!

Explore common interests

Senior professors don't have tenure pressure and may be open to various explorations.



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Senior professors don't have tenure pressure and may be open to various explorations.

Work closely with your advisor to find common interests
⇒ so that they can provide their best support.

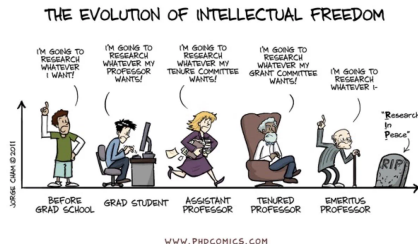


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Before asking a technical question, do the following:

Show enough prior work that others can build on what you already learned.

Before asking a technical question, do the following:

- Try to find an answer by searching the archives of the forum or mailing list you plan to post to.
- Try to find an answer by searching the Web.
- Try to find an answer by reading the manual.
- Try to find an answer by reading a FAQ.
- Try to find an answer by inspection or experimentation.
- Try to find an answer by asking a skilled friend.
- If you're a programmer, try to find an answer by reading the source code.

Show enough prior work that others can build on what you already learned.

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Choose a Channel That Can Respond

Documentation Established behavior, configuration, and supported usage.

Issue tracker Reproducible bugs tied to a specific project or version.

Public forum General technical questions that can help future readers.

Lab channel Project context, fast coordination, and preliminary diagnosis.

Direct message Private data, sensitive feedback, or a clearly responsible person.

Match the audience, scope, privacy, and expected response.

Make the Question Scannable and Reproducible

Subject

Use **object + deviation**:

“Two-GPU training — OOM after 600 steps”

Body

- 1 goal and expected behavior,
- 2 observed symptoms,
- 3 environment and reproduction steps,
- 4 evidence and attempted diagnosis,
- 5 one explicit question or request.

Clear + Correct + Complete + Concrete + Concise

Case: Replace a Guess with Diagnosable Evidence

Hard to diagnose

“Training crashes after a while. I think CUDA is broken. How can I fix it?”

Actionable evidence

- one GPU completes; two GPUs fail after about 600 steps,
- expected memory per GPU to decrease,
- batch-size reduction avoids the failure,
- changing data workers does not,
- command, memory trace, and error log attached.

Ask: Which measurement would distinguish duplicated model state from a data-pipeline leak?

Audit the Message Before You Send It

Clear Can the reader identify the purpose, conclusion, and ask?

Correct Have I verified the facts, numbers, labels, and links?

Complete Is there enough context to understand or reproduce the reasoning?

Concrete What evidence can the reader inspect?

Concise What can move to a link, appendix, or follow-up?

Courteous Does the wording preserve a productive working relationship?

Considered Is the abstraction level right, and does the logic flow?

A useful message enables the next decision or action.

Table of Contents

- 1 A General Guide
- 2 How to Communicate With Your Collaborator?
- 3 How to Ask Questions The Smart Way (From CS Perspective)?
- 4 **How to Do Presentation**

Presentation skills for computer science! (next lecture)

Thanks & Question Time!